

C4-2c: Basic Query

Basic SQL - Using DB Browser for SQLite



Download the **original database** from Coursemology again if you have modified it.

Execute SQL

Using SELECT

- View the data in all the fields in the table

```
SELECT * FROM Products;
```

- View the data in some of the fields in the table

```
SELECT Id, ProductName, UnitPrice FROM Products;
```

- View distinct data in a field

```
SELECT DISTINCT SupplierName FROM Products;
```



Execute SQL

Using WHERE

- Select all the customers from 'Mexico':

```
SELECT * FROM Customers WHERE Country = 'Mexico';
```

Operator	Description
=	Equal
>	Greater than
<	Lesser than
>=	Greater than or equal
<=	Lesser than or equal
<>	Not equal



Execute SQL

Using WHERE with other Operators

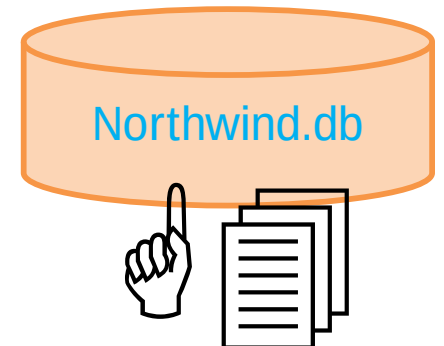
Operator	Description	Example
BETWEEN	Between a certain range	SELECT ProductName, UnitPrice FROM Products WHERE UnitPrice BETWEEN 10 AND 20;
LIKE	Search for a pattern	SELECT FirstName FROM Customers WHERE FirstName LIKE '%ja%'
IN	Multiple OR conditions	SELECT * FROM Customers WHERE Country IN ('Germany', 'France', 'UK')

Wildcard
Character

Wildcard Characters **%** and **_**:

bl% finds bl, black, blue and blur

h_t finds hot, hat and hit



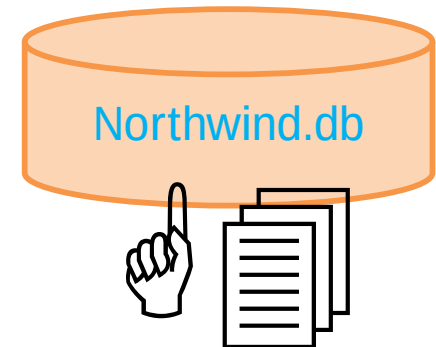
Execute SQL

Using WHERE with AND, OR, and NOT Operators

Operator	Example
AND	SELECT * FROM Customers WHERE Country='Germany' AND City='Berlin'
OR	SELECT * FROM Customers WHERE Country='Germany' OR Country='Spain'
NOT	SELECT * FROM Customers WHERE NOT Country='Germany'

Combining AND, OR, and NOT:

```
SELECT * FROM Customers
WHERE Country='USA' AND (City='Seattle' OR City='Portland');
```



Execute SQL

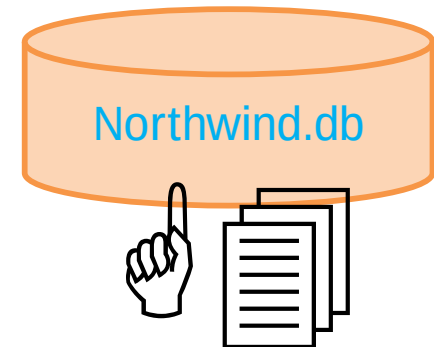
Using ORDER BY and LIMIT

```
SELECT ProductName, UnitPrice FROM Products  
ORDER BY UnitPrice DESC LIMIT 10;
```

Top 10 most
expensive
products

```
SELECT FirstName, Country FROM Customers  
ORDER BY Country ASC, FirstName DESC;
```

**The ORDER BY sorts the records
in ascending order by default.**



Execute SQL

Using GROUP BY

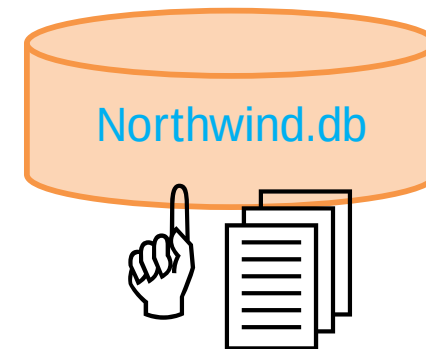
Execute the following SQL statements, one by one, and observe the outputs:

```
SELECT ProductName, SupplierCountry FROM Products;
```

```
SELECT ProductName, SupplierCountry FROM Products  
GROUP BY SupplierCountry;
```

```
SELECT COUNT(ProductName), SupplierCountry FROM Products  
GROUP BY SupplierCountry;
```

```
SELECT COUNT(ProductName), SupplierCountry FROM Products  
GROUP BY SupplierCountry  
ORDER BY COUNT(ProductName) DESC;
```



COUNT is an
Aggregate
Function

Execute SQL

Aggregate Functions

- COUNT, AVG, SUM, MAX, MIN

```
SELECT COUNT(Id) FROM Products;
```

```
SELECT AVG(UnitPrice) FROM Products;
```

```
SELECT SUM(Quantity) FROM Orders;
```

```
SELECT MAX(UnitPrice) FROM Products;
```

```
SELECT MIN(UnitPrice) FROM Products;
```



Execute SQL

Using HAVING

```
SELECT COUNT(Id), Country FROM Customers  
GROUP BY Country  
HAVING COUNT(Id) > 5  
ORDER BY COUNT(Id) DESC;
```

Note: SQL WHERE clause cannot be used with aggregate functions.

OUTPUT:

COUNT(Id)	Country
13	USA
11	France
11	Germany
9	Brazil
7	UK

